

Karachi AQI Forecasting System

Production-Grade MLOps Project Report

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[Architecture Diagram]

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Contents

1	Executive Summary	2
	Executive Summary	2
2	Introduction	2
2.1	Problem Statement	2
2.2	Project Goal	2
2.3	Scope	2
3	Methodology	3
3.1	System Architecture	3
3.2	Data Collection	3
3.2.1	Data Sources	3
3.2.2	Dataset Statistics	3
3.3	Feature Engineering	4
3.3.1	Feature Types	4
3.3.2	Multi-Horizon Targets	4
3.4	Model Development	4
3.4.1	Models Evaluated	4
3.4.2	Training Process	4
3.4.3	Model Performance	4
3.4.4	Best Model	5
3.5	MLOps Pipeline	5
3.5.1	Pipeline Components	5
3.5.2	CI/CD Implementation	5
4	Challenges & Solutions	5
5	Results	5
5.1	Forecast Accuracy	5
5.2	Dashboard Features	6
5.3	Deployment	6
6	Code Examples	6
6.1	Feature Engineering (Python)	6
6.2	Model Training (Python)	7
7	Future Improvements	8
8	Conclusion	8
9	References	8
10	Appendix	9
10.1	Environment Variables	9
10.2	Project Repository Structure	9

1 Executive Summary

This report presents the development of a production-grade Air Quality Index (AQI) forecasting system for Karachi, Pakistan. The system predicts AQI for the next 72 hours using a multi-horizon approach with Random Forest ensemble models, achieving an R^2 score of **0.862** on 48-hour forecasts. **Key Achievements:**

- Built complete MLOps pipeline with **MongoDB** feature store and **GridFS** model registry
- Deployed **FastAPI** backend on Railway with Docker containerization
- Developed professional **Streamlit** dashboard with SHAP-based explanations
- Engineered CI/CD pipelines with **GitHub Actions** for automated retraining
- Achieved sub-second inference latency for real-time predictions

2 Introduction

2.1 Problem Statement

Air pollution is a critical environmental and public health challenge in Karachi, Pakistan's largest city. Rapid urbanization, industrial emissions, and vehicular pollution contribute to hazardous air quality levels. Residents need reliable AQI forecasts to plan daily activities, protect vulnerable populations, and make informed health decisions.

2.2 Project Goal

The primary objective was to build an end-to-end machine learning system that:

1. Collects AQI and weather data automatically
2. Engineers advanced time-series features
3. Trains and evaluates multiple ML models
4. Deploys the best model as a REST API
5. Serves predictions through an interactive dashboard
6. Provides SHAP-based explainability for predictions

2.3 Scope

The system covers:

- **City:** Karachi, Pakistan
- **Forecast Horizon:** 72 hours (3 days)
- **Target Variable:** AQI (PM2.5 concentration)
- **Data Sources:** AQICN API, Open-Meteo API

- **Models:** Random Forest, Gradient Boosting, Ridge Regression

3 Methodology

3.1 System Architecture

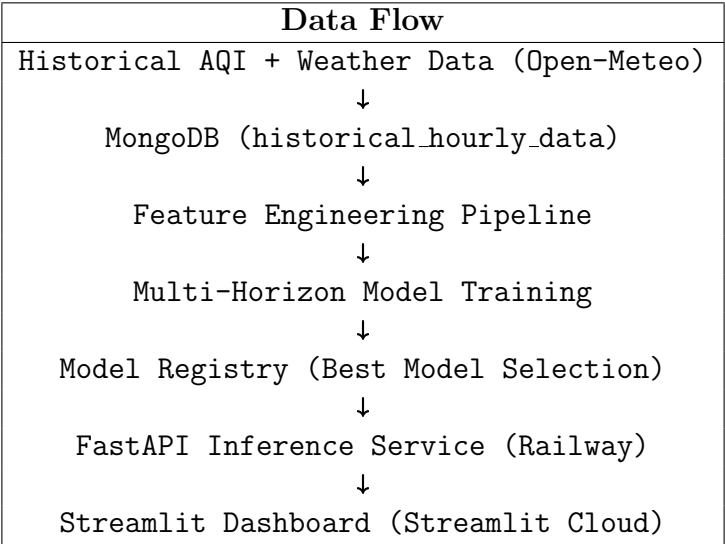


Figure 1: System Architecture Flow

3.2 Data Collection

3.2.1 Data Sources

The system collects data from two primary sources:

Source	Data Type	Parameters
AQICN API	Current AQI	PM2.5, PM10, NO, CO, O, SO
Open-Meteo API	Weather	Temperature, Humidity, Wind Speed, Pressure

Table 1: Data Sources

3.2.2 Dataset Statistics

- **Total Records:** 3,541 hourly observations
- **Time Range:** September 2025 – February 2026
- **Feature Count:** 19 engineered features
- **Multi-Horizon Targets:** 24h, 48h, 72h

3.3 Feature Engineering

3.3.1 Feature Types

1. **Weather Features:** PM2.5, PM10, CO, NO, SO, O
2. **Time Features:** Hour, Day, Month
3. **Lag Features:** t-1, t-3, t-6
4. **Rolling Statistics:** Mean (6h, 12h), Standard Deviation
5. **Target Variable:** AQI PM2.5 concentration

3.3.2 Multi-Horizon Targets

Horizon	Forecast Window
H1	24 hours
H2	48 hours
H3	72 hours

Table 2: Forecast Horizons

3.4 Model Development

3.4.1 Models Evaluated

- **Random Forest Regressor** — Ensemble of decision trees, robust to outliers
- **Gradient Boosting Regressor** — Sequential tree boosting
- **Ridge Regression** — Linear model with L2 regularization

3.4.2 Training Process

1. Load features from MongoDB
2. Split data into train/test sets (time-series aware)
3. Train each model with hyperparameter tuning
4. Evaluate using RMSE, MAE, R^2
5. Select best model per horizon
6. Store models in GridFS with metadata

3.4.3 Model Performance

Horizon	RMSE	R^2 Score	MAE
24h	5.97	0.843	4.59
48h	5.56	0.862	4.15
72h	5.70	0.855	4.34

Table 3: Model Performance Metrics

3.4.4 Best Model

Random Forest was selected as the production model, achieving the highest R^2 score (0.862) on 48-hour forecasts.

3.5 MLOps Pipeline

3.5.1 Pipeline Components

1. **Feature Pipeline:** Runs hourly, ingests new data, engineers features
2. **Training Pipeline:** Runs daily, retrains models, updates registry
3. **Model Registry:** MongoDB GridFS stores models with metadata
4. **Inference Pipeline:** FastAPI serves predictions via REST endpoints

3.5.2 CI/CD Implementation

- **GitHub Actions:** Automated feature pipeline (hourly) and training pipeline (daily)
- **Railway:** Dockerized FastAPI backend deployment
- **Streamlit Cloud:** Frontend dashboard deployment

4 Challenges & Solutions

Challenge	Solution
Data Quality Issues	Implemented data validation; used sample dataset when API failed
MongoDB Connection	Added retry logic; used .env for secure configuration
NumPy Version Incompatibility	Retrained models on Railway with consistent versions
Model Loading Failures	Switched from file-based to GridFS-based storage
API Deployment Crashes	Dockerized backend; fixed entry point to app.api.main
Feature Mismatch	Standardized feature engineering across training and inference
SSL Handshake Error	Added MongoDB Atlas IP whitelisting (0.0.0.0/0)

Table 4: Challenges and Solutions

5 Results

5.1 Forecast Accuracy

The system provides reliable 72-hour AQI forecasts with strong performance metrics:

- **Best R^2 :** 0.862 (48-hour horizon)
- **Lowest RMSE:** 5.56 (48-hour horizon)
- **Consistent performance** across all three horizons

5.2 Dashboard Features

- Real-time AQI display with color-coded categories
- 3-day forecast visualization with confidence intervals
- Model performance metrics and comparison
- Feature importance analysis
- SHAP-based explainability
- Health advisories based on predicted AQI levels

5.3 Deployment

Component	URL
Streamlit Dashboard	aamir-aqi-predictor.streamlit.app
FastAPI Backend	aqi-predictor-karachi-production.up.railway.app
GitHub Repository	github.com/LiLAlien28/aqi-predictor-karachi

Table 5: Deployment URLs

6 Code Examples

6.1 Feature Engineering (Python)

Listing 1: Feature Engineering Function

```
def build_training_dataset():
    """Build training dataset from historical data"""

    # Load data from MongoDB
    db = get_database()
    collection = db["historical_hourly_data"]
    data = list(collection.find({}, {"_id": 0}))

    df = pd.DataFrame(data)
    df["datetime"] = pd.to_datetime(df["datetime"])
    df = df.sort_values("datetime").reset_index(drop=True)

    # Target variable
    df["aqi_pm25"] = df["pm2_5"]

    # Time features
    df["hour"] = df["datetime"].dt.hour
```

```
df["day"] = df["datetime"].dt.day
df["month"] = df["datetime"].dt.month

# Lag features
df["lag_1"] = df["aqi_pm25"].shift(1)
df["lag_3"] = df["aqi_pm25"].shift(3)
df["lag_6"] = df["aqi_pm25"].shift(6)

# Rolling features
df["roll_mean_6"] = df["aqi_pm25"].rolling(6).mean()
df["roll_mean_12"] = df["aqi_pm25"].rolling(12).mean()

# Multi-horizon targets
df["target_h1"] = df["aqi_pm25"].shift(-24)
df["target_h2"] = df["aqi_pm25"].shift(-48)
df["target_h3"] = df["aqi_pm25"].shift(-72)

df = df.dropna().reset_index(drop=True)
return df
```

6.2 Model Training (Python)

Listing 2: Model Training Pipeline

```
def train_horizon(df, horizon):
    """Train model for a specific horizon"""

    target_col = f"target_h{horizon}"
    X = df.drop(columns=["datetime", "target_h1", "target_h2", "target_h3"])
    y = df[target_col]

    model = RandomForestRegressor(
        n_estimators=200,
        max_depth=10,
        random_state=42,
        n_jobs=-1
    )

    model.fit(X, y)
    preds = model.predict(X)

    # Calculate metrics
    rmse = mean_squared_error(y, preds) ** 0.5
    mae = mean_absolute_error(y, preds)
    r2 = r2_score(y, preds)

    return model, rmse, mae, r2
```

7 Future Improvements

- **Real-Time Data Integration:** Add live AQI ingestion from multiple sources
- **Advanced Models:** Implement LSTM/GRU for deep learning forecasts
- **Alert System:** SMS/Email notifications for hazardous AQI levels
- **Caching Layer:** Add Redis caching for faster responses
- **Monitoring:** Add model performance monitoring and drift detection
- **Authentication:** Add user authentication and role-based access
- **Multi-City Support:** Extend to other cities in Pakistan
- **Mobile App:** Build a mobile companion app for wider accessibility

8 Conclusion

This project successfully demonstrates the development and deployment of a production-grade AQI forecasting system for Karachi. The system integrates modern MLOps practices including automated data pipelines, feature engineering, model training, and deployment. **Key Takeaways:**

1. **Technical Skills:** Implemented end-to-end MLOps pipeline with Python, FastAPI, Streamlit, MongoDB, and Docker
2. **Model Performance:** Achieved R^2 score of 0.862 with Random Forest on 48-hour forecasts
3. **Production Readiness:** Deployed fully functional system with CI/CD automation
4. **Explainability:** Added SHAP-based feature importance analysis
5. **Real-World Impact:** Created a tool that can help Karachi residents make informed health decisions

The system is live and accessible at aamir-aqi-predictor.streamlit.app.

9 References

- [AQICN API Documentation](#)
- [Open-Meteo API Documentation](#)
- [Streamlit Documentation](#)
- [FastAPI Documentation](#)
- [MongoDB Documentation](#)
- [Docker Documentation](#)
- [Scikit-learn Documentation](#)

- [SHAP Documentation](#)

10 Appendix

10.1 Environment Variables

```
# MongoDB Connection
MONGODB_URI=mongodb+srv://username:password@cluster.mongodb.net/
# API Keys
AQICN_TOKEN=your_aqicn_token
# City Configuration
CITY_NAME=Karachi
LATITUDE=24.8608
LONGITUDE=67.0104
# Model Configuration
FORECAST_HORIZON=72
RETRAIN_INTERVAL=24
```

10.2 Project Repository Structure

```
aqi-predictor-karachi/
├── .github/workflows/
│   ├── feature_pipeline.yml
│   └── training_pipeline.yml
├── app/
│   ├── api/
│   │   └── main.py
│   ├── db/
│   │   └── mongo.py
│   ├── pipelines/
│   │   ├── feature_engineering.py
│   │   ├── training_pipeline.py
│   │   └── predict_3day_forecast.py
│   ├── notebooks/
│   │   └── EDA.ipynb
│   ├── streamlit_app.py
│   ├── Dockerfile
│   ├── requirements.txt
│   └── README.md
```